

11 (long)

(a) Sketch how the logarithm of crack growth per cycle under cyclic loading da/dN , varies with the logarithm of the stress intensity factor range ΔK . On your diagram, indicate the threshold ΔK_{th} , the Paris regime and the behaviour as K_{max} approaches the fracture toughness K_{IC} . [6]

(b) The fuselage of a prototype commercial jet aircraft has a radius R of 1.3 m. The aircraft is used typically for an average of six short-haul flights each day. During each flight, the cabin is pressurised to a pressure P of 50 kPa. The fuselage skin thickness t is 0.9 mm ($t \ll R$) and is made of an aluminium alloy with $K_{IC} = 30 \text{ MPa m}^{1/2}$. In laboratory tests, the alloy exhibited the following crack growth rates:

$$\begin{aligned} \frac{da}{dN} &= 10^{-8} \text{ m cycle}^{-1} & \text{at} & \quad \Delta K = 5 \text{ MPa m}^{1/2} \\ \frac{da}{dN} &= 10^{-7} \text{ m cycle}^{-1} & \text{at} & \quad \Delta K = 8.9 \text{ MPa m}^{1/2} \end{aligned}$$

(i) The aircraft fuselage and air pressurisation system are designed such that the cabin pressure can be sustained with a through-thickness crack. Determine the critical length of a through-thickness longitudinal crack of length $2a$ perpendicular to the hoop direction at the flight pressure. Assume that the dimensionless constant Y , in the expression for the stress intensity factor K , is equal to 1. [4]

(ii) Consider a through-thickness crack 75 mm long. Will fast fracture occur before there is a drop in cabin pressure? [4]

(iii) Assuming that the fatigue crack growth rate for the material satisfies Paris' Law, determine the constants n and A in Paris' Law (Materials Databook). [6]

(iv) The aircraft is subjected to a major maintenance check every 2 years. When a routine check was conducted on the aircraft, the presence of a 2 mm long through-thickness longitudinal crack in the fuselage was left undetected due to human error. Using the constants determined in part (iii), determine the remaining life of the aircraft following the maintenance check. Is the aircraft likely to fail or suffer a loss of cabin pressure before the next major maintenance check? [10]